

MATH 2790 Fall 2025

Homework 10

Due date: Wednesday, Nov 12

Instructions:

- (a) It is expected to show detailed work for every problem. You are allowed to have a discussion with peers on some challenging questions. But then you should work on your own to finish the assignment.
- (b) All questions in this assignment are from sections 5.2 and 5.4

Problems:

1. Question (1) on page 256 of textbook under section 5.2.
2. Question (2) on page 256 of textbook under section 5.2.
3. Question (5) on page 257 of textbook under section 5.2. (Hint: correct the monthly interest rate to 5 decimal places as a percentage. When finding the payment, round up to the nearest cent, since the final payment is slightly reduced.)
4. Question (1) on page 259 of textbook under section 5.4.
5. Question (2) on page 259 of textbook under section 5.4.
6. Kevin takes out a 10-year loan of L , which he repays by the amortization method at an annual effective interest rate of i . Kevin makes payments of 1000 at the end of each year. The total amount of interest repaid during the life of the loan is also equal to L . Calculate the amount of interest repaid during the first year of the loan.